

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (E&E) (CBCS) Degree Examinations
 December 2023

EE2101-1 – ANALOG SIGNAL PROCESSING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- If the op-amp in the Fig 1. has an input offset voltage of 5mV and the open loop voltage gain of 10000, the V_o will be
 A) +50V
 B) -50V
 C) $\pm 15V$
 D) +15V
- Inverting op-amp with open loop gain of 10^5 is shown in Fig. 2. The closed loop gain $\frac{V_o}{V_s}$ is
 A) -8
 B) -11
 C) -10
 D) -9

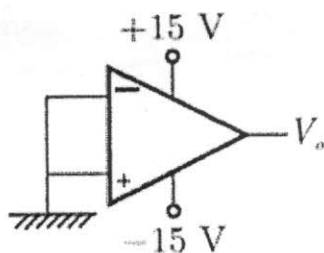


Fig. 1.

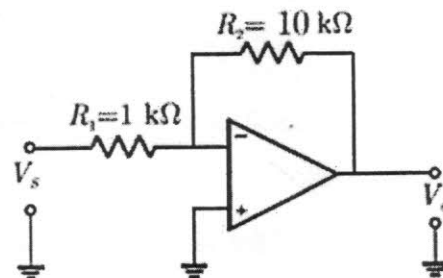


Fig. 2.

- The frequency at which gain is 0db for integrator is (R_1 is the input resistance, R_F is the feedback resistance, C_F is the feedback capacitance)
 A) $f=1/(2\pi R_F C_F)$
 B) $f=1/(2\pi R_1 C_F)$
 C) $f=(1/2\pi) \times (R_F/R_1)$
 D) $f=1/(R_1 C_F)$
- The factor that makes the differentiator circuit unstable is _____
 A) Output impedance
 B) Input voltage
 C) Noise
 D) Gain
- The hysteresis voltage for the inverting Schmitt trigger circuit with following specification: $R_2 = 56k\Omega$, $R_1 = 100\Omega$, $V_{ref} = 0V$ & $V_{sat} = \pm 14V$.
 A) 0mV
 B) 25mA
 C) -25mA
 D) 50mA
- If $V_{GS} \gg 0$ then N-channel Enhancement type MOSFET will be in _____ state and P-channel Enhancement type MOSFET will be in _____ state
 A) ON, ON
 B) ON, OFF
 C) OFF, ON
 D) OFF, OFF
- If an NMOS is degenerated by a resistor in series with the source, what will happen to the output resistance?
 A) It increases
 B) It decreases
 C) It remains same
 D) Cannot be determined
- N-MOSFET is also called?
 A) IGFET
 B) IGBT
 C) PFET
 D) NFET

9. In which of the following region does MOSFET acts as an amplifier?
 A) Ohmic B) Cut-off
 C) Saturation D) Non-saturation
10. The input impedance of a common Gate MOSFET ____?
 A) is high B) is low
 C) depends on external components D) varies with gate voltage
11. For a Common Gate amplifier the input resistance R_{in}
 A) $R_{in} < \frac{1}{g_m}$ B) $R_{in} = \frac{1}{g_m}$
 C) $R_{in} > \frac{1}{g_m}$ D) $R_{in} = g_m$
12. Which of the following type of effect does a MOSFET amplifier uses?
 A) Field effect B) Polarity effect
 C) Charge effect D) Capacitance effect
13. In the low frequency magnitude response of a CS amplifier there are ____ breakpoints corresponding to ____ poles
 A) 2, 2 B) 3, 2
 C) 3, 3 D) 4, 3
14. Body transconductance is defined as
 A) $g_{mb} = \frac{i_D}{v_{GS}} \Big|_{v_{BS}, v_{DS} = \text{constant}}$ B) $g_{mb} = \frac{\partial i_D}{\partial v_{BS}} \Big|_{v_{GS}, v_{DS} = \text{constant}}$
 C) $g_{mb} = \frac{i_D}{\partial v_{GS}} \Big|_{v_{BS}, v_{DS} = \text{constant}}$ D) $g_{mb} = \frac{\partial i_D}{\partial v_{GS}} \Big|_{v_{BS}, v_{DS} = \text{constant}}$
15. The gate capacitance $C_{gs} = C_{ds} = \frac{1}{2} WLC_{ox}$ when the MOSFET is operating in ____ region
 A) Saturation B) Triode
 C) Cutoff D) all
16. The unity gain frequency of MOSFET is given by
 A) $f_T = \frac{g_m}{2\pi[C_{gs} + C_{gd}]}$ B) $f_T = \frac{[C_{gs} + C_{gd}]}{2\pi g_m}$
 C) $f_T = \frac{g_m}{\pi[C_{gs} + C_{gd}]}$ D) $f_T = \frac{g_m}{[C_{gs} + C_{gd}]}$
17. The depletion NMOS will operate in the triode region as long as ____ voltage does not exceed the ____ voltage by more than $|V_t|$
 A) drain, gate B) gate, drain
 C) source, gate D) gate, source
18. The input impedance ____ and output impedance ____ with feedback in a Series - Shunt feedback amplifier configuration.
 A) increases increases B) decreases decreases
 C) increases decreases D) decreases increases
19. To obtain large CMRR for a differential amplifier the bias current source should have a ____ output resistance
 A) Low B) High
 C) Constant D) variable
20. The fall off rate of high frequency response beyond the upper cut-off frequency is
 A) -20dB/decade B) -40dB/decade
 C) -60dB/decade D) 0dB/decade

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the following terms with reference to op-amps. (i) Input offset voltage (ii) Input offset current	04	L*1	1	1,2
	b) A capacitor coupled voltage follower using 741 op-amp is to have an input impedance of $12\text{k}\Omega$. The lower cutoff frequency can have a maximum value of 100Hz and the load resistance can vary from $3.3\text{k}\Omega$ to $15\text{k}\Omega$. Design a suitable circuit.	06	L3	1	2,3
	c) Explain with neat cross-sectional view of the physical structure and working of the enhancement-type NMOS transistor.	06	L2	2	2,3
2.	a) With a neat circuit diagram, and design steps, explain the working of capacitor coupled inverting amplifier circuit. Show how to set the lower and upper cut-off frequencies?	06	L2	1	2,3
	b) Explain the working of Zener diode peak clipper circuit.	04	L1	1	1,2
	c) Draw the I_D - V_{DS} characteristics of MOSFET. Identify various regions of Operation of the Enhancement NMOS Transistor.	06	L2	2	2,3
3.	a) Explain the working of a non-saturating precision full wave precision rectifier using half wave rectifier and summing amplifier. Discuss the design procedure to be followed.	06	L2	1	1,2
	b) Explain the parameters to be considered while designing op-amp in switching circuits.	03	L4	1	2,3
	c) Consider a process technology for which $L_{\min} = 0.4\text{ }\mu\text{m}$, $t_{\text{ox}} = 8\text{ nm}$, $\mu_n = 450\text{ cm}^2/\text{V}\cdot\text{s}$, and $V_t = 0.7\text{ V}$. (i) Find C_{ox} and k'_n . (ii) For a MOSFET with $W/L = 8\text{ }\mu\text{m}/0.8\text{ }\mu\text{m}$, calculate the values of V_{OV} , V_{GS} , and V_{DSmin} needed to operate the transistor in the saturation region with a dc current $I_D = 100\text{ }\mu\text{A}$. (iii) For the device in (ii), find the values of V_{OV} and V_{GS} required to cause the device to operate as a 1000Ω resistor for very small V_{DS} .	07	L3	2	2,3

Unit – II

4. a) Calculate the drain current of M_1 in Fig. 4(a) if $\mu_n C_{\text{ox}} = 100\text{ }\mu\text{A}/\text{V}^2$, $V_{\text{TH}} = 0.5\text{ V}$, and $\lambda = 0$. What value of R_D is necessary to reduce I_D by a factor of 2?

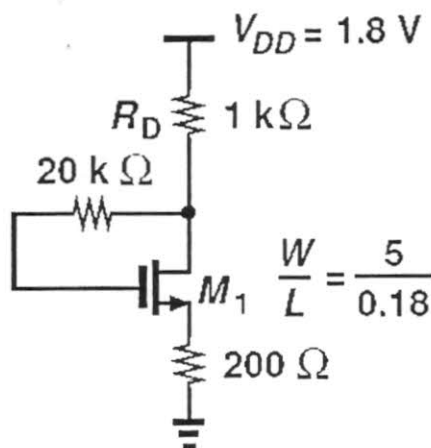


Fig. 4(a)

- | | | | | | |
|----|---|----|----|---|-----|
| b) | With a neat circuit diagram, explain the CS amplifier configuration with source resistance. | 06 | L3 | 3 | 2,3 |
| c) | Explain the CMOS inverter topology. Discuss its overall input/output characteristic showing various regions of operation. | 05 | L2 | 4 | 1,2 |
| | | 05 | L2 | 4 | 2,3 |

5. a) It is required to design the circuit shown in Fig 5(a) to establish a dc drain current $I_D = 0.5\text{mA}$. The MOSFET has following specifications: $V_t = 1\text{V}$ and $k'_n W/L = 1\text{mA/V}^2$. Assume $\lambda = 0$. Use a power-supply $V_{DD} = 12\text{V}$. Calculate the percentage change in the value of I_D obtained when the MOSFET is replaced with another unit having the same but $V_t = 1.5\text{V}$.

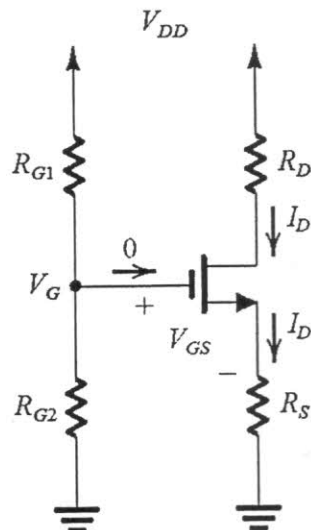


Fig. 5(a)

- | | | | | |
|--|----|----|---|-----|
| b) Develop the T equivalent-circuit model for the MOSFET with r_o . | 07 | L3 | 3 | 2,3 |
| c) Differentiate between common drain amplifier and op-amp voltage follower. | 06 | L2 | 4 | 2,3 |
| | 03 | L2 | 3 | 2,3 |
6. a) Define transconductance. Mention the three relationships to determine transconductance.
- b) With neat equivalent circuit discuss the high frequency model for the MOSFET.
- c) A discrete common-source MOSFET amplifier utilizing a drain-to-gate feedback resistance $R_G = 10\text{M}\Omega$ for biasing purposes. The drain resistance $R_D = 10\text{k}\Omega$. The input signal v_i is coupled to the gate via a large capacitor, and the output signal at the drain is coupled to the load resistance R_L of $10\text{k}\Omega$ via another large capacitor. Determine the transconductance and small-signal voltage gain of the amplifier circuit.
- | | | | | |
|--|----|----|---|-----|
| | 04 | L1 | 3 | 1,2 |
| | 05 | L2 | 4 | 2,3 |
| | 07 | L4 | 4 | 2,3 |

Unit – III

- | | | | | |
|--|----|----|---|-----|
| 7. a) Explain the complete three frequency band response of the CS amplifier. | 08 | L1 | 5 | 1,2 |
| b) Explain the MOS differential pair operation with a common mode input voltage. | 08 | L2 | 5 | 2,3 |
8. a) Prove that the minimum closed loop gain should be larger than 29 for a FET based RC phase shift oscillator. Explain the operation with neat circuit diagram and equivalent circuit.
- b) Explain the Miller theorem with an example.
- | | | | | |
|--|----|----|---|-----|
| | 10 | L4 | 5 | 2,3 |
| | 06 | L1 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
